

MANJEET PATHAK

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EDUCATION

Indian Institute of Information Technology Bhopal
B.Tech in Information Technology

Dec 2021 – June 2025
CGPA: 9.66

WORK EXPERIENCE

Juspay | Haskell, Fintech, Rust, Docker, Envoy
ASDE

Aug 2025 – Present
Full Time

- Designed and implemented a **globally distributed active-active** multi-cell architecture using AWS Global Accelerator and multi-region load balancing, aiming to achieve **99.995% uptime** with seamless failover
- Engineered a **routing-id-based affinity system (x-routing-id)** and built an **ID replication pipeline (Redis Streams)** to maintain routing consistency and availability across distributed cells
- Built a multi-layered request resolution system (**routing affinity + replicated lookup + fanout fallback**) and optimized Istio service mesh routing, significantly improving fault tolerance and latency
- Built observability and control systems (**health checks, maker-checker, Rust-based dashboard**) for real-time monitoring, traffic management, and merchant onboarding
- Contributed to core **Haskell** backend services and resolved critical production issues, ensuring system reliability

D+A Strategies | Python, Backtesting, Quant Analysis, Algo Trading
Quant Analyst Associate

May 2026 – Present
Part Time

- Delivered lectures on quant research pitfalls — including **survivorship bias, look-ahead bias, overfitting, and backtest design** — to educate and guide less experienced division members
- Developing an advanced trend-following strategy for the analyst team focused on quantitative signal generation, adaptive modeling, and systematic risk management

Buyhatke Internet Pvt Ltd | NodeJS, MySQL, ExpressJS, CRON
SDE - Backend

Oct 2024 – July 2025
Full Time

- Built a **payment routing system** across multiple payment gateways, ensuring reliable transaction processing based on scores
- Integrated a **gift card system** with multiple partners via **Onramp.money**, enabling secure transactions
- Developed backend systems using Express.js and MySQL, optimizing queries and automating workflows; improved **SpendLens** and **AutoCoupons** to support **50K+ daily users**

ACHIEVEMENTS

- Ranked **108th out of 228** in **ACM ICPC finals (Amritapuri)** – 2023
- Qualified Round-2 of **HackerCup 2023 and 2024** with a best rank of **1875**
- Expert** at **Codeforces** (**maxRating – 1605**) with global rank **736** in Round 1024
- 5 Star** at **CodeChef** (**maxRating – 2033**) with best global rank of **32** at Starters 176
- Guardian** at **LeetCode** (**maxRating – 2170**) — top **1.13%** globally
- Awarded **1st Runner-Up** in National Level Hackathon **CodeUtsava 6.0**
- Department Topper** Information Technology IIIT Bhopal (2021–2025)

PROJECTS

Low-Latency Order Book & Matching Engine | C++23, libcurl, Google Benchmark, CMake

[GitHub](#)

- Built a limit order book in modern C++23 achieving **~22M ops/sec on aggressive sweeps** and **~9M ops/sec on inserts (46–108 ns/op)**, benchmarked with Google Benchmark on Apple M-series
- Engineered a **cache-padded wait-free SPSC ring buffer**, **object-pool allocator**, and **intrusive FIFO per price level** for zero heap allocation on the hot path; integrated live Binance Futures/Spot feed with end-to-end event→fill latency telemetry (**p50 ~10–20 μs**)

Event-Driven Trading Engine | Rust, Tokio, Async Traits

[GitHub](#)

- Built a high-performance **event-driven trading engine** in **Rust** with async multi-exchange market data ingestion and pluggable strategy architecture across **Binance, Bybit, CoinGecko, and Binance Futures**
- Developed modular quant infrastructure with **paper-trade execution, portfolio management, realized/unrealized PnL tracking**, and cross-exchange arbitrage support
- Engineered a configurable risk-management system with **exposure controls, position sizing, daily-loss limits, trade validation, and automated kill-switch protection**

SKILLS

Languages: C++ (C++23), Python, Rust, Haskell, C, JavaScript, TypeScript, SQL

Quant & Trading: Algorithmic Trading, Backtesting, Order Book Systems, Market Microstructure, Trend Following, Mean Reversion

Systems & Concepts: Low-Latency Systems, Lock-Free Data Structures, SPSC Queues, Distributed Systems, Service Mesh (Istio/Envoy)

Frameworks & Libraries: Tokio, Serde, libcurl, Google Benchmark, Express, NodeJS, NextJS, React

Databases: PostgreSQL, MySQL, MongoDB, Redis

Dev Tools: Git, Docker, CMake, Cargo, Linux, GDB